

Real-Time Air-Quality Monitoring Platform with Personalized Recommendations

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Introduction

- Ambient and household air pollution causes ~7 million premature deaths annually; 99% of people breathe air that exceeds WHO guidelines.
- Latin-American megacities such as Bogotá still record PM_{2.5} hot-spots despite mitigation plans.
- Public APIs (AQICN, Google Air Quality, IQAir) expose rich data streams, yet lack personalization and unified access.

Goal

- Research Question:** How can we deliver highly-available, low-latency air-quality insights that adapt to user context in real time?
- Objective:** Design and validate a PostgreSQL-centric architecture that ingests multi-source AQI streams every 10 min and produces color-coded health recommendations within 2 s (p95).

Proposed Solution

- Ingestor:** Python micro-service polls three public APIs; raw JSON archived in MinIO for audit.
- Normalizer:** Aligns units scales, writes to TimescaleDB hypertable partitioned by city × month.
- Query Acceleration:** Concurrently-refreshed materialised views feed dashboards API.
- Recommendation Engine:** Maps EPA AQI bands + user profile to actionable advice product tips.
- Interfaces:** REST/GraphQL endpoints, Grafana dashboards, mobile push alerts.

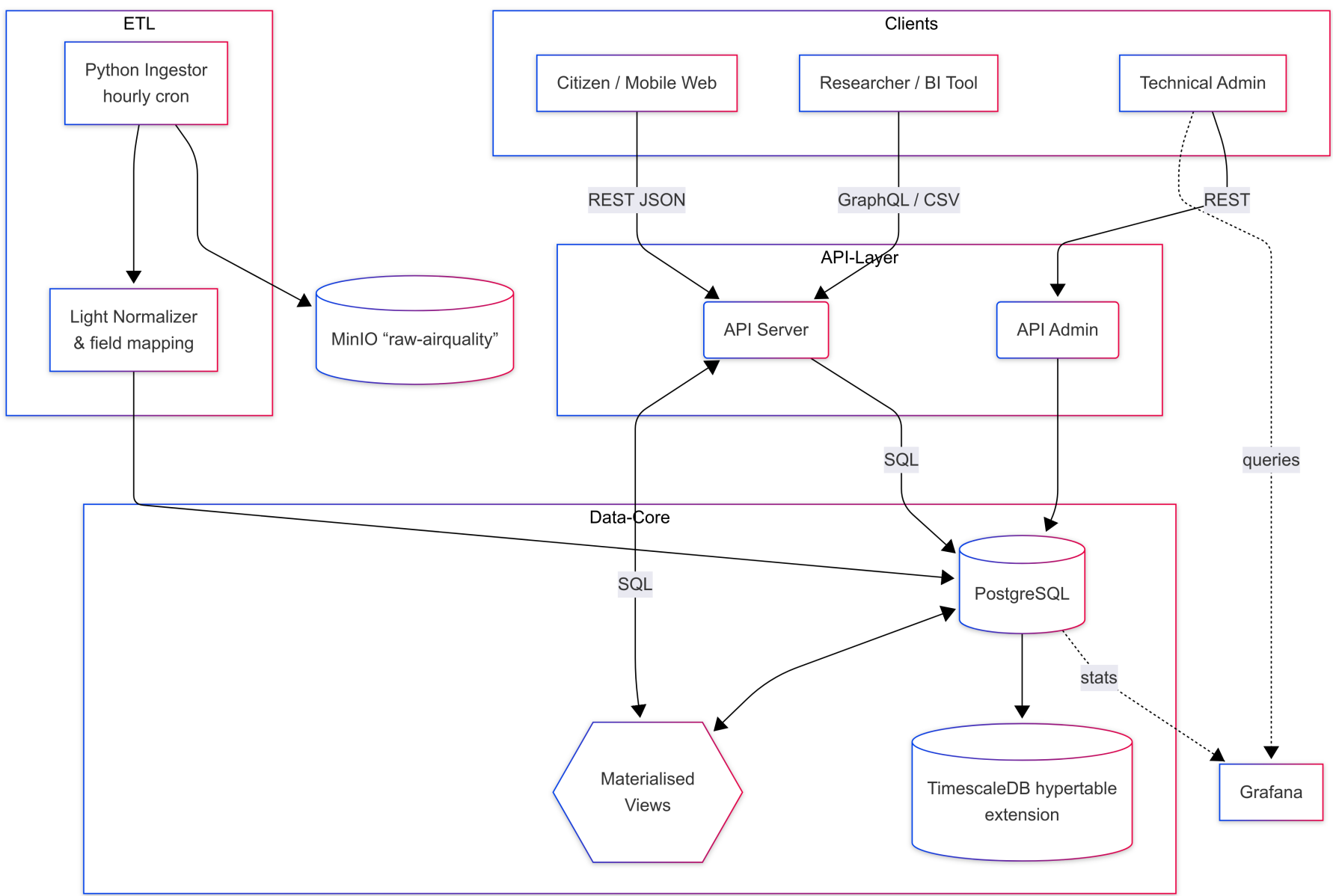


Figure 1: End-to-end architecture (planned).

Experimental Setup

- Dataset:** AQICN historical CSV (2015-2024). Phase-1 loads 2022-2024 Bogotá subset (~30M rows).
- Hardware:** 4-vCPU/16 GB primary DB + two hot read replicas; NVMe storage.
- Load Test:** Apache JMeter emulates 1 000 concurrent users, 5 requests/s each for 10 min.
- Metrics:** ingest lag, dashboard latency, materialised-view refresh, CPU, recommendation accuracy.

Expected Results

Metric	Target p95	Status
Ingest lag	≤2 s	TBD
Dashboard latency	≤2 s	TBD
Materialized-view refresh	≤5 s	TBD
Recommendation accuracy	≥90%	TBD

Benchmarks will be updated after Phase-1 deployment.

References

References

- [1] World Health Organization, "Global Air Quality Guidelines", 2021.
- [2] Timescale Inc., "Tables and Hypertables", Docs, 2025.
- [3] AQICN Project, "Real-Time Air Quality API", 2024.

Conclusions Future Work

- A PostgreSQL+TimescaleDB+MinIO stack can meet sub-2 s end-to-end latency for 1 000 concurrent users without Kafka/Flink overhead.
- Materialised views and partitioning keep query costs bounded as the dataset scales.
- Next steps: ingest edge sensor networks, add LSTM forecasts, deploy multi-region logical replication.